

👉 XAI: Interactive Explanations — XAL and TalkToModel

FAMAF - UNC

First Semester 2026



Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

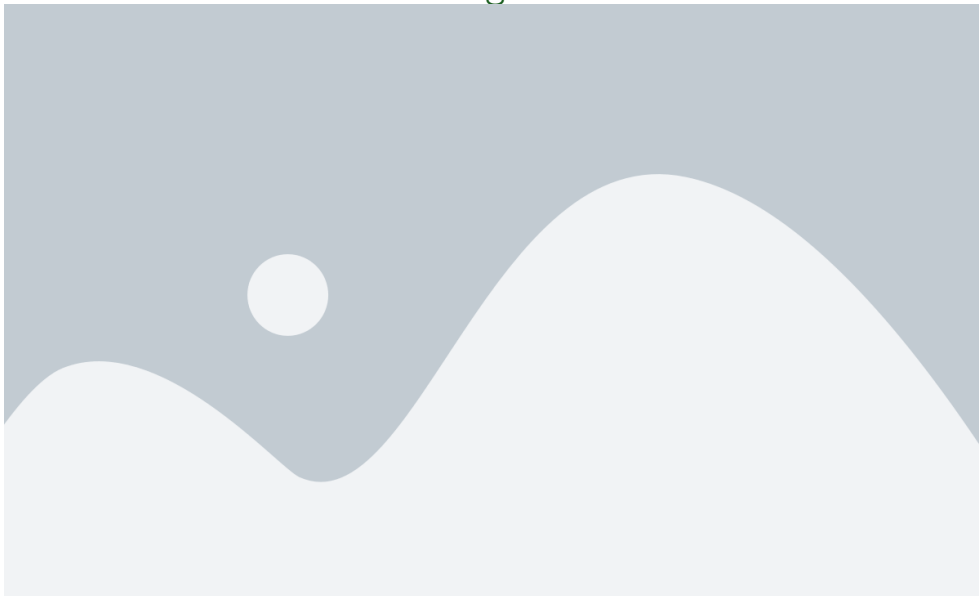
🍷 Paper 1: Explainable Active Learning (XAL) ---

Explainable Active Learning (XAL): Toward AI Explanations as Interfaces for Machine Teachers

Ghai, Moon, Kumar, Cheng, Zhang — CSCW 2021

(Ghai et al. 2021)

🍷 Motivation: Active Learning Bottleneck ---



🍷 The XAL Paradigm ---

Explainable Active Learning

- 1 Model sends prediction + **explanation** to annotator
- 2 Annotator confirms/rejects + **provides feedback**
- 3 Model updates from annotation and feedback



🍷 Explanation Format ---



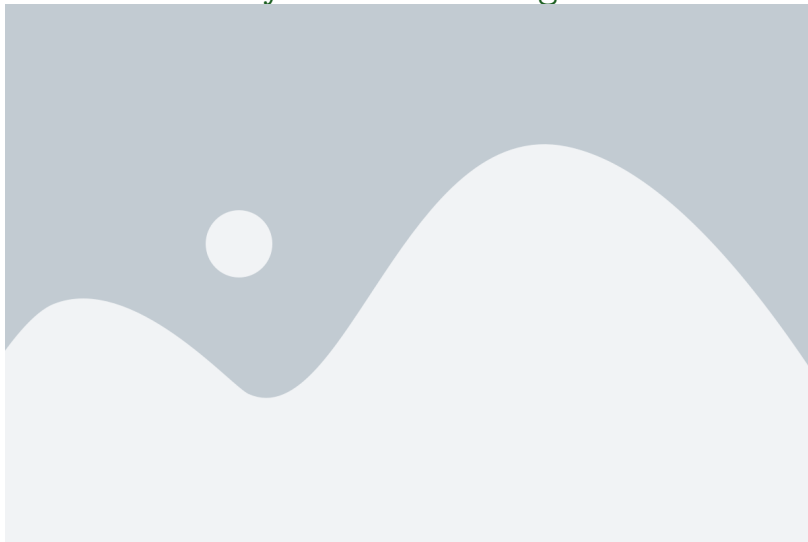
🍷 Task: Adult Income Classification ---

- Dataset: Adult Income (UCI)
- Task: Predict whether income $>$ \$80k (binary classification)
- Model: Logistic regression with L2 regularization
- Explanation method: local feature importance

🍷 Study Design ---



🍷 AL Simulation: Early vs. Late Stage ---



- Model accuracy curve defines **early stage** (low accuracy) and **late stage** (high accuracy)

Research Questions ---

RQ1: How does human vs. model accuracy change across stages?

RQ2: Does XAL affect user experience?

- H1: Trust calibration ✓
- H2: Satisfaction ×
- H3: Cognitive workload ✓

RQ3: What moderates XAL effects?

- H4: Task knowledge ×
- H5: AI experience ✓
- H6: Need for cognition ✓

RQ4: What determines explanation quality ratings?

🍷 Key Findings: Trust Calibration ---

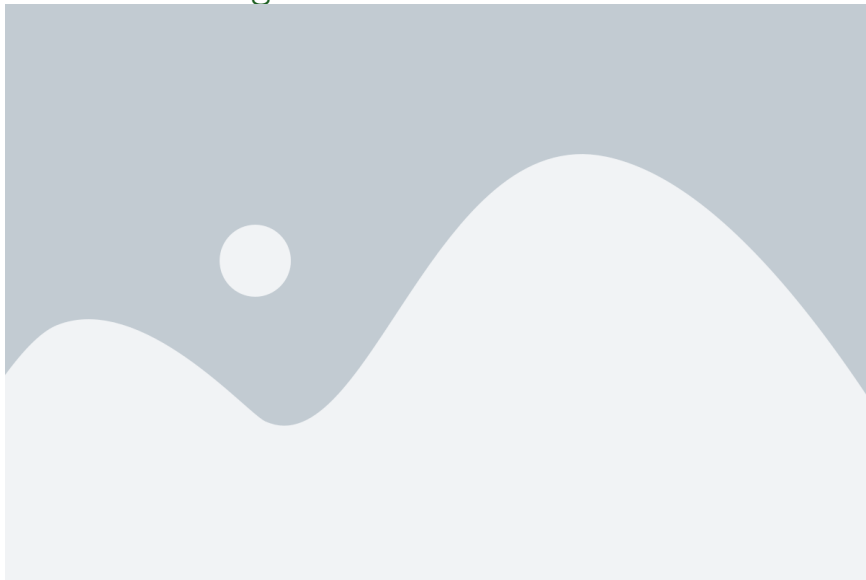


- **H1 confirmed:** XAL users show **low trust early** (model is wrong), **increasing trust late** (model improves)

🍷 Key Findings: Explanation Ratings ---

- Explanation ratings are **higher** when the model is correct
- Explanation ratings are **higher** when annotators **wrongly disagree** with the model in the late stage
- Suggests users judge explanations by outcome, not reasoning quality

🍷 Open Feedback Categories ---



XAL Conclusions ---

Contributions

- First study combining XAI with Active Learning
- XAL improves **trust calibration** and reduces **cognitive overload**
- Explanations enable annotators to act as "machine teachers"

Limitations

- Small N (36 AMT)
- Only one explanation method
- Interpretable model only (LR)

Future: mitigate anchoring (Bućinca et al., CSCW'21), learn from explanation feedback

XAL Discussion Questions ---

- ➊ In which scenarios would XAL be most/least useful?
- ➋ How can we prevent anchoring on incorrect explanations?
- ➌ Would XAL work with less interpretable models?

Explaining Machine Learning Models with Interactive Natural Language Conversations Using TalkToModel

Slack, Krishna, Lakkaraju, Singh — Nature Machine Intelligence 2023

(Slack et al. 2023)

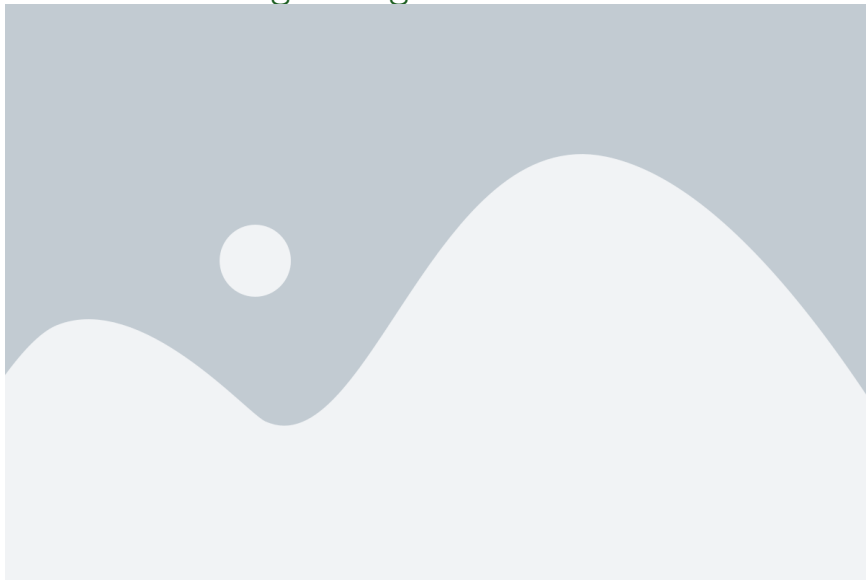
🍷 Motivation: The Explainability Bottleneck ---



🍷 System Overview ---



🍷 Component 1: Dialogue Engine ---

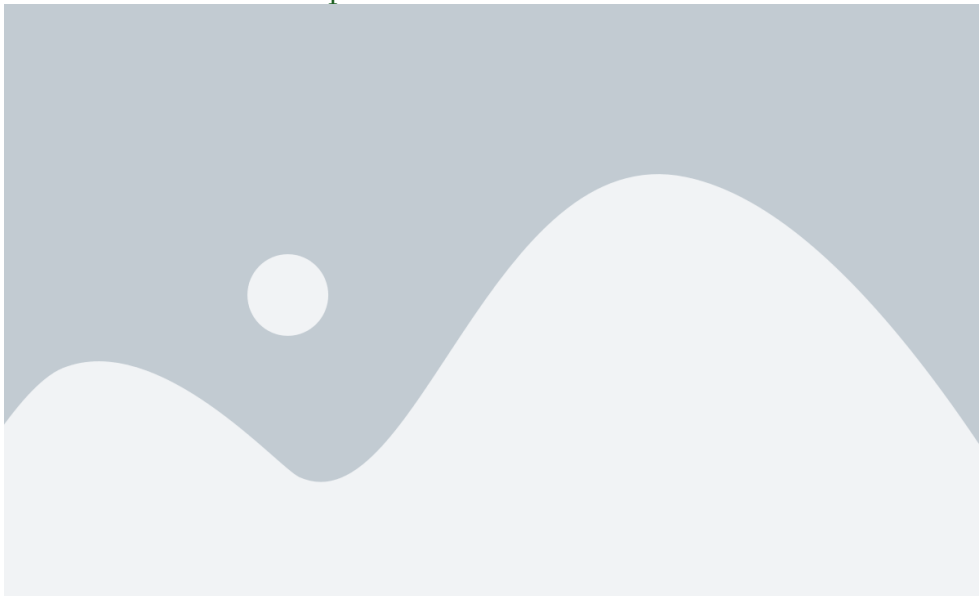


Grammar-based domain-specific language

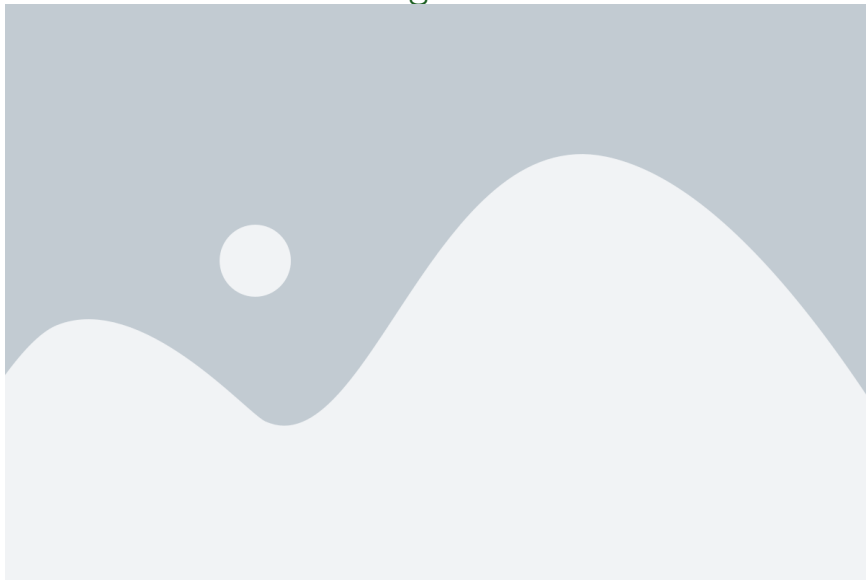
Fine-Tuning Data Generation ---

- Authors write **50 (utterance, parse) pairs** per domain
 - ▶ Every operation appears ≥ 2 times
- MTurk: paraphrase each utterance **8 ways** = 400 pairs
- MTurk: rate fidelity of paraphrase (keep $\geq 3/4$ averaged over 5 raters)
- Manual filtering by authors
- Enumerate wildcards $\rightarrow$ **20k–40k training pairs** per domain

🍷 The Grammar: DSL Operations ---



🍷 Component 2: Execution Engine ---



🍷 Faith and Fudge Scores ---



Fudge score (faithfulness of a mask $\mathbf{m}$ for instance $\mathbf{x}$):

🍷 Experiment 1: LLM Evaluation ---

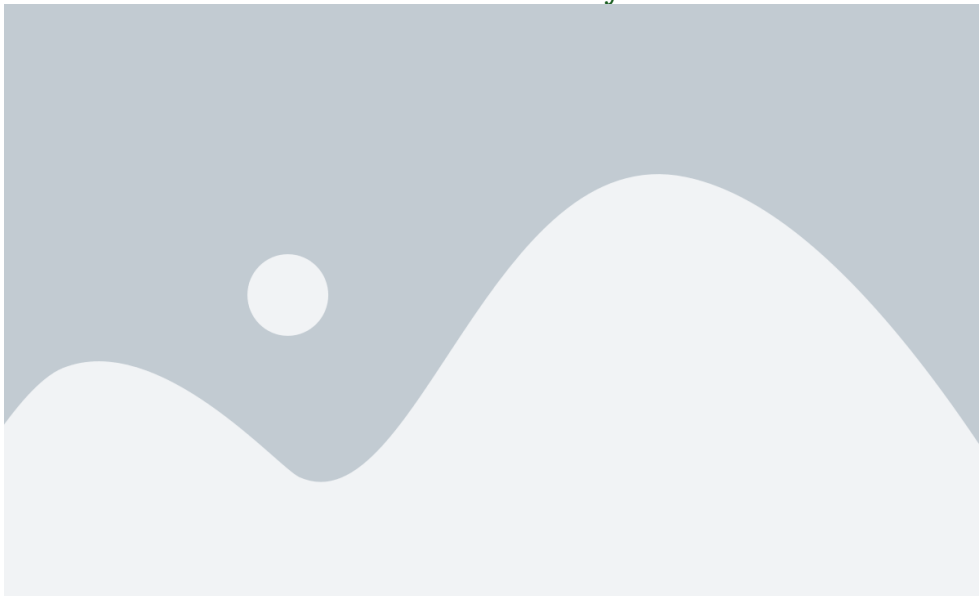
Datasets:

- Diabetes (Pima Indian): 768 women, 8 features, 400→190 questions
- Credit (German): 1000 applicants, 20 features, 400→200 questions
- Recidivism (COMPAS): 11757 defendants, 43 features, 400→146 questions

Splits:

- **IID (Easy)**: operations seen in training, different arguments
- **Compositional (Hard)**: new operation combinations not seen in training

🍷 LLM Results: Exact Match Accuracy ---



Experiment 2: Grammar Coverage ---

Question: Is the grammar expressive enough to capture all XAI questions?

Method

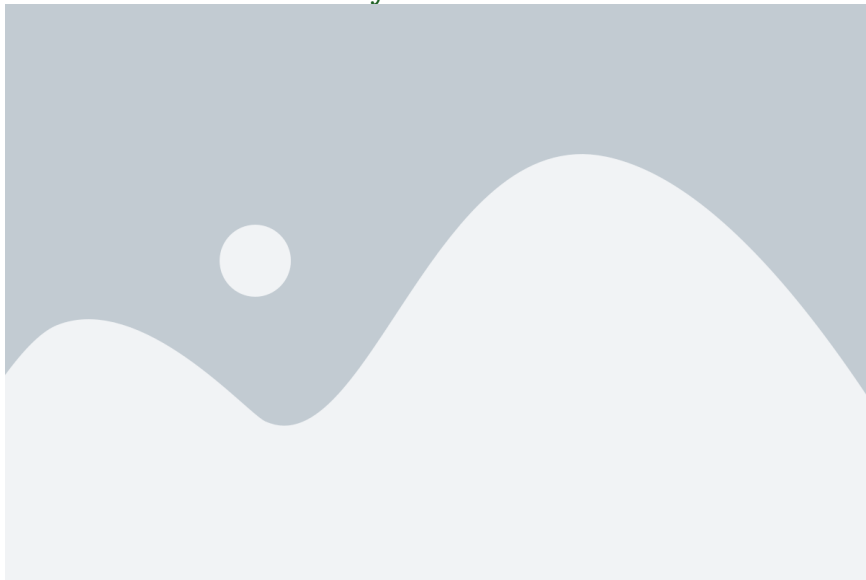
- Use a curated **XAI question bank** (31 questions, informed by expert interviews)
- Manually review if grammar operations can answer each

Result

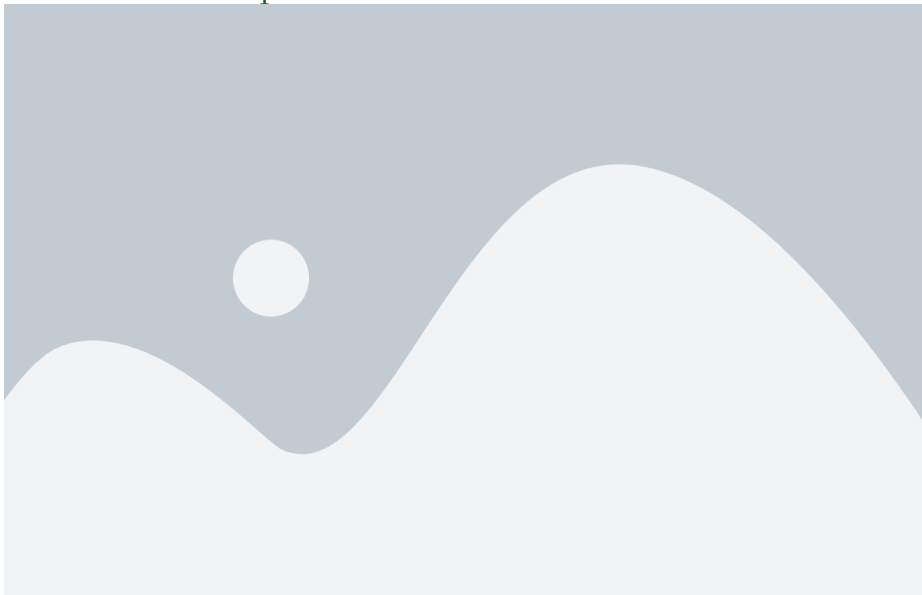
30/31 questions can be answered by the grammar

Example: `topk(test_data, all), mistakes(test_data), cfe(filter(test_data, id, A, =), 10, Q)`

🍷 Experiment 3: User Study ---



🍷 The Baseline: `explainerdashboard ---`



🍷 TalkToModel Interface ---



User Study Findings ---

TalkToModel vs. explainerdashboard

- **Over 90% accurate** vs. $\sim 60\%$ with explainerdashboard
- Answered questions in **half the time**
- **Consistently preferred** on ease of use, confidence, speed, and likability

% Questions Completed

Healthcare workers: 86.2% vs. 74.7%

ML grads: 93.9% vs. 73.8%

% Accuracy on Completed

Healthcare workers: 91.8% vs. 66.1%

ML grads: 100% vs. 62.5%

TalkToModel Conclusions ---

- **Elegant UI**: natural language makes model interpretation accessible to laypeople and ML practitioners
- **Highly extensible**: handles a variety of XAI methods, problem domains
- **Reasonably accurate**: fine-tuned T5 interprets user intent well
- **Low barrier**: only need your own dataset to deploy

TalkToModel Limitations ---

Limitations

- Not tested in **real-world settings**
- **No flexibility** in explanation method selection ("most feasible CFE", "most stable explanation")
- **No guarantees on data quality** for fine-tuning; some manual labor required
- **No domain knowledge** beyond the grammar
- Accuracy on **compositional (hard) split** still very low

TalkToModel Discussion Questions ---

- 1 Are ML practitioners most responsible for the accessibility of XAI?
- 2 Is the TalkToModel LLM itself interpretable? When is it acceptable to improve XAI with more black-box AI?
- 3 How much control over explanations should we give users while remaining accessible to laypeople?
- 4 Does TalkToModel's existence excuse other XAI methods from being accessible?
- 5 Dashboard vs. Dialogue — which is better and when?

Thank You!

- Ghai, Bhavya, Q Vera Moon, Abhay Kumar, Yongfeng Cheng, and Kai Zhang. 2021. “Explainable Active Learning (Xal): Toward Ai Explanations as Interfaces for Machine Teachers.” *Proceedings of the ACM on Human-Computer Interaction* 4 (CSCW3): 1–28.
- Slack, Dylan, Satyapriya Krishna, Himabindu Lakkaraju, and Sameer Singh. 2023. “Explaining Machine Learning Models with Interactive Natural Language Conversations Using Talktomodel.” *Nature Machine Intelligence* 5 (8): 873–83.